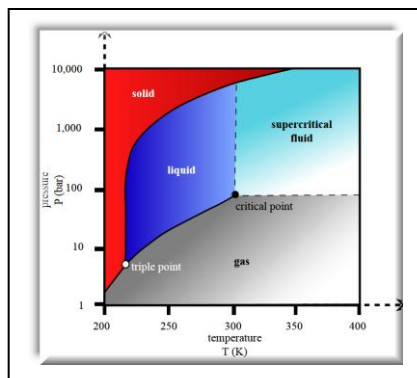




College:
Department:
Course:
Classroom and Time:

UIC COLLEGE OF ENGINEERING
UIC Department of Chemical Engineering
CHE 301: Chemical Engineering Thermodynamics
EIB 271 M|W: 1:00 PM – 2:15 PM

Instructor:
Office hours:
E-mail:
Cell Phone:

Dr. Michael Caracotsios, EIB 244
M|Tu|W|Th: Stop by my office or schedule a ZOOM meeting
mcaracot@uic.edu
(630) 696-0934

TA Name:
E-mail:
Office hours:
Office Location:

Ece Bayraktar
ebayr@uic.edu
Th: 02:00PM - 04:00PM
EIB 247

COURSE OBJECTIVES

This course is designed to extend the student's acquired knowledge and understanding of the first and second law of thermodynamics to the most critical areas of Chemical Engineering Thermodynamics and Chemical Reaction Equilibrium. These areas encompass properties of pure fluids and fluids in homogeneous solutions, single and multicomponent phase equilibrium, and chemical equilibrium. The knowledge of this material will provide a strong foundation for all subsequent courses in chemical engineering and contribute towards understanding of chemical and physical processes. By the end of the course the students should be able to:

- ✚ Understand the basic concepts of the five thermodynamic potentials: Internal Energy, Gibbs and Helmholtz Free Energy, Entropy and Enthalpy.
- ✚ Apply control volume analysis to a variety of situations related the phase and chemical equilibrium and solve problems utilizing computer software.
- ✚ Understand the theory of thermodynamics of ideal and non-ideal solutions
- ✚ Understand the applicability and use of various Equations of State in describing the thermodynamic state of pure fluids and mixtures thereof.
- ✚ Understand the applicability and use of various Liquid Activity Coefficient Methods in describing the thermodynamic state of non-ideal liquid mixtures
- ✚ Determine equilibrium conditions between phases using equations of state and liquid activity coefficients.
- ✚ Determine equilibrium conditions between reactants and products using equilibrium constants and Gibbs energy minimization concepts.

COURSE PREREQUISITES

The following are key requirements for a successful completion of this course.

- **Mathematics:** Differential and Integral Calculus
- **Statistics:** Linear and Nonlinear Regression
- Numerical Methods for Chemical Engineering
- **Programming Experience:** Excel VBA, Matlab, C++
- **CHE Department Prerequisites:** CHE 201, CHE 205 and CHE 210

REQUIRED SOFTWARE

All students must download and install the Microsoft EXCEL [32-bit Version] **CHE Calculator v13.0** following the instructions listed on the UIC Blackboard in the **Computational Tools** content area of the course.

REQUIRED TEXTBOOK

Michael Caracotsios, *An Introduction to Chemical Engineering Thermodynamics. Theory and Applications*. First Edition, 2026 will be distributed to all students free of charge in the **Course Textbook** content area of the CHE 301 course.

RECOMMENDED TEXTBOOKS

1. Themis Matsoukas, *Fundamentals of Chemical Engineering Thermodynamics*, Prentice Hall, 2013.
2. J. M. Smith, H. C. Van Ness and M. M. Abbott, *Introduction to Chemical Engineering Thermodynamics*, 7th Edition, McGraw-Hill, 2005.
3. S. I. Sandler, *Chemical and Engineering Thermodynamics*, 3rd Edition, John Wiley & Sons, Inc., 1999.

CHE 301: COURSE OUTLINE**1. CHAPTER 1: REVIEW**

- Mass and Energy Balances.
- Momentum and Mechanical Energy Balances.
- Entropy and Lost Work Balances.
- Calculus and Numerical Methods for Chemical Engineers

2. CHAPTER 2: APPLICATIONS OF MACROSCOPIC BALANCES

- Adiabatic Mixing
- Thermodynamic Cycles for Power Production
- Other Applications from Chemical Engineering.

3. CHAPTER 3: THERMAL PROPERTIES OF PURE FLUIDS

- Thermodynamic Potentials
- Review: Maxwell Equations
- Thermal Properties of Pure Fluids: Ideal Gases and Liquids
- Vapor-Liquid Equilibrium between Ideal Gases and Liquids

4. CHAPTER 4: EQUATIONS OF STATE FOR PURE FLUIDS

- Cubic and Virial Equations of State
- Residual Properties for Enthalpy, Entropy and Heat Capacity
- Fugacity and Fugacity Coefficients from Equations of State
- Thermal Properties of Pure Fluids with Equations of State
- Vapor-Liquid Equilibrium Fundamentals using Equations of State

5. CHAPTER 5: THERMAL PROPERTIES OF MIXTURES

- Partial Molar Properties
- Thermal Properties of Ideal Gas and Liquid Mixtures
- Gibbs Thermodynamic Consistency Theorem
- Vapor Liquid Equilibrium for Ideal Gas and Liquid Mixtures
- Bubble point, Dew Point and Pressure-Temperature Flash Calculations

6. CHAPTER 6: EQUATIONS OF STATE FOR MIXTURES

- Cubic Equations of State and Mixing Rules
- Thermal Properties of Non-ideal Gas and Liquid Mixtures
- Fugacity and Fugacity Coefficients of Components in Mixtures
- VLE Calculations for Non-Ideal Gas and Liquid Mixtures
- Bubble point, Dew Point and Pressure-Temperature Flash Calculations

7. CHAPTER 7: LIQUID ACTIVITY COEFFICIENTS

- Gibbs Generating Excess Functions
- Liquid Activity Coefficients (NRTL, van Laar, Wilson, Margules)
- Bubble Point, Dew Point and Pressure-Temperature Flash Calculations
- Azeotrope Finding Calculations
- VLE Data Reduction Methods and Thermodynamic Consistency Tests

8. CHAPTER 8: CHEMICAL REACTION EQUILIBRIUM

- Criteria for Equilibrium in Chemical Reactions
- Gibbs Free Energy and Equilibrium Constants
- Evaluation of Equilibrium Constants from Thermodynamic Data
- Calculation of Equilibrium Composition for Single and Multi-Reaction Systems

CHE 301: COURSE GRADING AND OTHER POLICIES

Your final grade for the **CHE 301** Chemical Engineering Thermodynamics course will be determined from (**max 10**) take home assignments, two midterms and one final examination, and (**max 10**) quiz/attendance records. The contribution of all these activities to the grade calculation formula is shown below:

1. Home Assignments: [40%]
2. Examinations: [40%]
3. Quizzes & Attendance: [20%]

The formula for the calculation of your total score is given by the equation below. The grades associated to a total score range are also given in the table below.

$$TS(\text{Total Score}) = 0.40\langle H \rangle + 0.40\langle E \rangle + 0.20\langle Q + A \rangle$$

Total Score	Grade	Symbol	Max Score	Meaning of Symbols
$85 \leq TS \leq 100$	A			
$70 \leq TS < 85$	B	$\langle H \rangle$	100 pts	Average Score for Home Assignments
$60 \leq TS < 70$	C	$\langle E \rangle$	100 pts	Average Score for Examinations
$50 \leq TS < 60$	D	$\langle Q + A \rangle$	100 pts	Average Score for Quizzes & Attendance

Note 1: Policy of Missed or Late Work:

Completion of all Assignments is mandatory. In the event that a student does not complete an Assignment by the designated time and date, the student will receive a score of zero points. The student may still be able to negotiate a better score by turning in the missed assignment at a time agreed upon with the Professor. If attendance is taken and a student is absent the student will receive a score of zero points, unless there is a reasonable explanation for the absence. Please email the Professor if you face an unexpected situation that may impede your attendance.

Note 2: Partial and Extra Credit Policy:

Partial credit can be requested under certain circumstances. The student should make an appointment with the Professor and be prepared to discuss the reasons for requesting the extra credit. Final decision rests with the Professor.

Extra credit may be given at the discretion of the Professor. The extra credit will be based on the student's record of attendance, punctuality, participation in class, good work and ethics behavior, and reasonable scores on all assignments and examinations. The extra credit will be contingent on perfect attendance and completion of all home assignments.

Note 3: Academic Integrity Policy:

As a student and member of the UIC community, you are expected to adhere to the Community Standards of academic integrity, accountability, and respect. Please review the UIC Student Disciplinary Policy for additional information.

Note 4: E-mail Expectations:

Students are responsible for all information instructors send to your UIC email and Blackboard accounts. Faculty messages should be regularly monitored and read in a timely fashion.

Dr. Michael Caracotsios

CLINICAL ASSOCIATE PROFESSOR

CHEMICAL ENGINEERING DEPARTMENT

UNIVERSITY OF ILLINOIS CHICAGO

Email: mcaracot@uic.edu

MISCELLANEOUS

Blackboard Collaborate Help

By email: LTS@uic.edu

By phone: (312) 413-0003 Option #1

ACCOMMODATIONS

Disability Accommodation Procedures

UIC is committed to full inclusion and participation of people with disabilities in all aspects of university life. If you face or anticipate disability-related barriers while at UIC, please connect with the Disability Resource Center (DRC) at drc.uic.edu, via email at drc@uic.edu, or call (312) 413-2183 to create a plan for reasonable accommodations. To receive accommodations, you will need to disclose the disability to the DRC, complete an interactive registration process with the DRC, and provide me with a Letter of Accommodation (LOA). Upon receipt of an LOA, I will gladly work with you and the DRC to implement approved accommodations.

Religious Accommodations

Following [campus policy](#), if you wish to observe religious holidays, you must notify me by the tenth day of the semester. If the religious holiday is observed on or before the tenth day of the semester, you must notify me at least five days before you will be absent. Please submit [this form](#) by email with the subject heading: **"YOUR NAME: Requesting Religious Accommodation."**

CLASSROOM ENVIRONMENT

Inclusive Community

UIC values diversity and inclusion. Regardless of age, disability, ethnicity, race, gender, gender identity, sexual orientation, socioeconomic status, geographic background, religion, political ideology, language, or culture, we expect all members of this class to contribute to a respectful, welcoming, and inclusive environment for every other member of our class. If aspects of this course result in barriers to your inclusion, engagement, accurate assessment, or achievement, please notify me as soon as possible.

Name and Pronoun Use

If your name does not match the name on my class roster, please let me know as soon as possible. My pronouns are [\[she/her; he/him; they/them\]](#). I welcome your pronouns if you would like to share them with me. For more information about pronouns, see this page: <https://www.mypronouns.org/what-and-why>.

Community Agreement/Classroom Conduct Policy

- Be present by turning off cell phones and removing yourself from other distractions.
- Be respectful of the learning space and community. For example, no side conversations or unnecessary disruptions.
- Use preferred names and gender pronouns.
- Assume goodwill in all interactions, even in disagreement.
- Facilitate dialogue and value the free and safe exchange of ideas.
- Try not to make assumptions, have an open mind, seek to understand, and not judge.

- Approach discussion, challenges, and different perspectives as an opportunity to “think out loud,” learn something new, and understand the concepts or experiences that guide other people’s thinking.
- Debate the concepts, not the person.
- Be gracious and open to change when your ideas, arguments, or positions do not work or are proven wrong.
- Be willing to work together and share helpful study strategies.
- Be mindful of one another’s privacy, and do not invite outsiders into our classroom.

Content Notices and Trigger Warnings

Our classroom provides an open space for a critical and civil exchange of ideas, inclusive of a variety of perspectives and positions. Some readings and other content may expose you to ideas, subjects, or views that may challenge you, cause you discomfort, or recall past negative experiences or traumas. I intend to discuss all subjects with dignity and humanity, as well as with rigor and respect for scholarly inquiry. If you would like me to be aware of a specific topic of concern, please email or visit my Student Drop-In Hours.

RESOURCES: Academic Success, Wellness, and Safety

We all need the help and the support of our UIC community. Please visit my **drop-in hours** for course consultation and other academic or research topics. For additional assistance, please contact your assigned college advisor and visit the support services available to all UIC students.

Academic Success

- [UIC Tutoring Resources](#)
- College of Engineering [tutoring program](#)
- [Equity and Inclusion in Engineering Program](#)
- [UIC Library](#) and [UIC Library Research Guides](#).
- [Offices](#) supporting the UIC Undergraduate Experience and Academic Programs.
- [Student Guide for Information Technology](#)
- [First-at-LAS](#) Academic Success Program, focusing on LAS first-generation students.

Wellness

- **Counseling Services:** You may seek free and confidential services from the Counseling Center at <https://counseling.uic.edu/>.
- Access [U&I Care Program](#) for assistance with personal hardships.
- **Campus Advocacy Network:** Under Title IX, you have the right to an education that is free from any form of gender-based violence or discrimination. To make a report, email TitleIX@uic.edu. For more information or confidential victim services and advocacy, visit UIC’s Campus Advocacy Network at <http://can.uic.edu/>.

Safety

- [UIC Safe App](#)—PLEASE DOWNLOAD FOR YOUR SAFETY!
- [UIC Safety Tips and Resources](#)
- [Night Ride](#)
- [Emergency Communications](#): By dialing 5-5555 from a campus phone, you can summon the Police or Fire for any on-campus emergency. You may also set up the complete number, (312) 355-5555, on speed dial on your cell phone.